DATA ENGINEERING & ANALYTICS

Live Market Intelligence.

A Microsoft Fabric case study in real-time tick ingestion, Eventstream routing, KQL Database modelling, Real-Time Intelligence dashboards, and Activator alerts for an algorithmic trading firm replacing batch market data.

VelocityTrade Capital

Algorithmic trading · Real-Time Intelligence
Eventstream · KQL · Activator

MICROSOFT FABRIC

CASE STUDY 11 · CONTENTS

Six sections. Ten deliverables. One streaming pipeline.

An algorithmic trading firm. A 1.8 billion dollar book. 240 monitored symbols. Market data refreshed on a 15 to 45 minute batch cycle. The signals our models generate are only as good as the data they consume. The estimate is 2.1 to 3.4 million dollars per quarter in return leakage from timing errors. This is the brief.

01

Executive & Value at Stake

Return leakage in basis points per trade. Compliance reconstruction liability growing with every trade. The mandate is unambiguous.

PAGE 3

02

Industry Context & Diagnosis

Data infrastructure is alpha infrastructure. Three compounding failures: source fragmentation, no streaming enrichment, no event-driven alerting.

PAGE 4 · 5

03

Target Architecture & Schemas

Eventstream, Eventhouse, KQL Database, RTI dashboard, Activator alert. Every component named, sized, and bounded.

PAGE 6 · 7

04

Engineering Mandate & Query Library

Eight KQL queries. Three dashboard tiles. One Activator alert. The complete artefact set the Trading Domain will consume.

PAGE 8 · 9

05

Deliverables & Acceptance

Ten named deliverables. Seven acceptance criteria. The end-to-end trace from sample source to alert evaluation.

PAGE 10 · 11

06

Discussion & Lab

Eight design questions. Lab instructions phase by phase. Risk register.

PAGE 12 · 13

SECTION 01 · EXECUTIVE RECOMMENDATION

Close the gap between data and market. Before the next volatility event.

VelocityTrade Capital operates at the intersection of algorithmic trading and real-time data infrastructure, where the fidelity and latency of market data directly determines the quality of every automated trading decision. The current market data capability has reached a critical inflection point. Batch-refreshed prices, adequate during the firm's early growth, are now a structural liability that constrains every systematic strategy running across the trading book.

THE VALUE AT STAKE

Internal back-testing indicates that algorithmic signals generated against data lagging by twenty minutes or more produce entry and exit timing errors that erode strategy returns by twelve to eighteen basis points per trade cycle. Across an average monthly volume of 4,200 automated executions, this represents 2.1 to 3.4 million dollars per quarter of return leakage attributable directly to data infrastructure latency rather than strategy underperformance.

THE COMPLIANCE DIMENSION

Regulatory frameworks governing algorithmic trading require firms to demonstrate the ability to reconstruct the market data state at the precise moment any automated decision was made. The current batch model produces no event-level audit trail. Each refresh overwrites the prior state. In a regulatory inquiry, VelocityTrade cannot provide the granular reconstruction MiFID II requires.

THE MANDATE

Microsoft Fabric Real-Time Intelligence provides the platform foundation. Deploy a governed Eventstream pipeline that ingests live tick data, routes continuously enriched events into a KQL Database, and surfaces live signals through a Real-Time Intelligence dashboard backed by automated Activator alerts. The transition is not optional. It is the prerequisite for the firm's strategies to operate against data that reflects market reality.

> "The signals our models generate are only as good as the data they consume. We are making automated decisions in a market that has moved eighteen minutes since our last refresh. That is not a data engineering problem. That is a trading risk problem inside a data engineering wrapper."

HEAD OF QUANTITATIVE STRATEGY, VELOCITYTRADE CAPITAL

L

Latency

Fifteen to forty-five minute batch refresh cycles replaced by continuous event-driven tick ingestion.

I

Intelligence

Sub-second trading signals derived from live OHLCV tick events across all monitored symbols.

C

Compliance

Complete event-level audit trail. Reconstruction of market data state at any decision timestamp.

SECTION 02 · INDUSTRY & CLIENT CONTEXT

Data infrastructure is alpha infrastructure. The latency profile of the feed is the edge.

Algorithmic trading firms do not differentiate primarily on the quality of strategies in isolation. They differentiate on the speed and fidelity of the data that feeds those strategies. A well-constructed mean reversion model executed against live tick data will consistently outperform an identical model against twenty-minute-stale data. Not because the strategy is better, but because the signal is closer to ground truth.

THE FIRM AND THE BOOK

VelocityTrade Capital is a mid-size algorithmic trading firm focused on North American equity markets, primarily NYSE and NASDAQ. The firm runs systematic strategies spanning momentum, mean reversion, and statistical arbitrage across 240 actively monitored symbols. Execution is fully automated through a proprietary order management system that consumes signals from the quantitative research platform.

THE CRITICAL DEPENDENCY

Every signal, every risk threshold, and every position limit in the firm's operating model is anchored to a price reference. That reference currently arrives via a batch process refreshed on a fifteen to forty-five minute cycle. The technology stack has scaled three years of growth. The market data layer has not. There is a structural mismatch between the latency expectations of the strategies and the latency reality of the data.

EXHIBIT 1 · ORGANISATIONAL & DATA LANDSCAPE

DOMAIN	SYSTEM	PAIN POINT	TIER
Trade Execution	Proprietary OMS	Signals fed by stale batch prices.	CORE
Market Data	Batch pipeline	15 to 45 minute latency per cycle.	BLOCKER
Risk Management	Excel	False positives from stale thresholds. Manual rate inputs.	FRAGILE
Compliance Audit	None	No tick-level trail. Cannot reconstruct decision state.	EXPOSED
Quantitative Research	Notebook platform	Ad hoc tick queries with no governed access to live data.	UNGOVERNED

SECTION 02 · PROBLEM DIAGNOSIS

Three layers of compounding failure. Batch ingestion. No enrichment. No alerts.

The current state diagnosis reveals three compounding failure layers that together create a market data capability structurally unfit for the firm's operating model. Each layer would be a meaningful gap on its own. Together they form the structural liability.

01

Source Fragmentation

The batch pipeline samples the continuous stream at intervals, collapsing rich tick granularity into coarse OHLCV snapshots. Intra-period price paths are discarded entirely.

15-45m
Refresh cycle latency

02

No Streaming Enrichment

No mechanism to enrich incoming events before they land. Derived fields computed at query time against stale snapshots, pushing compute into the consumption layer.

0
Stream attributes pre-computed

03

No Event-Driven Alerting

Price movements detected only through manual dashboard monitoring. During high market activity, when attention is most divided, material moves can go undetected.

$3.4M
Quarterly return leakage

DIAGNOSTIC FINDING

The absence of a real-time tick ingestion and enrichment pipeline creates systematic market data latency of fifteen to forty-five minutes across all 240 monitored symbols, producing estimated return leakage of $2.1M to $3.4M per quarter attributable directly to entry and exit timing errors driven by stale price references. The complete absence of an event-level audit trail exposes the firm to MiFID II non-compliance, with reconstruction liability growing with every automated trade executed against a batch-sourced price snapshot.

WHAT THE VOLATILITY LOOKBACK HIDES

A stock that moved from 142.00 to 156.00 and back to 143.50 within a single 15-minute window appears in the current system as a negligible net movement. The volatility event that every real-time risk model would have flagged as a position review trigger is invisible to the firm. The strategies are blind to exactly the events they were designed to capture.

WHY THE ALERTING GAP MATTERS MOST

Activator provides the alert layer the firm needs but requires a governed, continuously updated KQL table as its foundation, which the current batch architecture cannot provide. The alerting gap is therefore an architectural consequence, not a configuration oversight. Fix the foundation and the alerts become available.

SECTION 03 · TARGET ARCHITECTURE

Every component is named. Every responsibility is bounded.

The target architecture is fully governed and event-driven, built entirely within Microsoft Fabric Real-Time Intelligence. The batch refresh model is replaced by a continuous ingestion layer that captures individual tick events as they occur, enriches them before they land in a queryable store, and makes them immediately available to trading signals, dashboard tiles, and automated alert conditions.

EXHIBIT 3 · TARGET ARCHITECTURE COMPONENTS

COMPONENT	TYPE	NAME	ROLE
Workspace	Fabric Workspace	VelocityTrade-RTI	ISOLATE
Eventhouse	Fabric Eventhouse	VelocityTradeEventhouse	HOST
Tick Table	KQL Table	StockTicks	PRIMARY
Audit Table	KQL Table	PipelineAuditLog	CHANGE LOG
Eventstream	Fabric Eventstream	TickIngestionStream	INGEST
Source	Sample Data	StockMarketSample	FEED
Dashboard	RTI Dashboard	VelocityTrade Live Market Intelligence	SURFACE
Alert	Fabric Activator	DrawdownAlert	NOTIFY
Query Library	KQL Queryset	MarketDataQueries	GOVERN

NON-NEGOTIABLE GOVERNANCE

All tick events enter the KQL Database exclusively through TickIngestionStream. No direct writes from notebooks or external tools are permitted at any stage. Write access is restricted to the Eventstream pipeline identity. All KQL queries used by dashboard tiles or trading signals must be validated against live data before deployment. Every configuration change is recorded in PipelineAuditLog with timestamp, author, component changed, and description.

SECTION 03 · TABLE SCHEMAS

Two tables. The tick store and the audit log.

The KQL Database hosts two tables. StockTicks is the primary store for all ingested OHLCV tick events. PipelineAuditLog records every configuration change made to any pipeline component across the full delivery. Both schemas must be defined and confirmed before any data lands.

EXHIBIT 4 · STOCKTICKS

FIELD	TYPE	DESCRIPTION	ROLE
`Ticker`	string	Stock ticker symbol identifying the traded instrument.	KEY
`Date`	datetime	Timestamp of the tick event in UTC.	TIME
`Open`	real	Opening price for the tick period.	OHLC
`High / Low`	real	Highest and lowest prices recorded during the tick period.	OHLC
`Close / AdjClose`	real	Closing price and adjusted close accounting for corporate actions.	OHLC
`Volume`	long	Total shares traded during the tick period. Long type required: NYSE and NASDAQ symbols can produce per-tick volumes exceeding the integer range.	MEASURE

EXHIBIT 5 · PIPELINEAUDITLOG

FIELD	TYPE	DESCRIPTION	ROLE
`ChangeTimestamp`	datetime	UTC timestamp of the configuration change.	WHEN
`Author`	string	Name of the engineer who made the change.	WHO
`ComponentChanged`	string	Name of the pipeline component that was modified.	WHAT
`ChangeDescription`	string	Description of what was changed and why.	WHY

SECTION 04 · ENGINEERING MANDATE

Critical trading infrastructure. Not a proof of concept.

You are the Lead Data Engineer reporting to the Enterprise Platform Owner. Your mandate is to design, provision, and validate the governed real-time market data pipeline that will serve as the firm's authoritative source of live tick intelligence. This is critical trading infrastructure that will underpin automated trading decisions, risk threshold calculations, and regulatory compliance.

NON-NEGOTIABLE MANDATES

Connect the StockMarketSample exclusively through TickIngestionStream. No direct table writes from notebooks at any stage. Define the StockTicks schema before any data lands. Restrict write access to the Eventstream pipeline identity. Record every configuration change in PipelineAuditLog with full metadata. Validate every query against live streaming data before registration. The complete pipeline must be traceable end to end from sample source to dashboard tile and alert evaluation.

EXHIBIT 7 · DASHBOARD TILES

TILE	QUERY	VISUAL	PURPOSE
Tile 1	Q1 Latest Close	Table	REFERENCE
Tile 2	Q2 Intra-Day Range	Bar Chart	VOLATILITY
Tile 3	Q4 Volume Surge	Table	ATTENTION

EXHIBIT 8 · DRAWDOWNALERT SPECIFICATION

ATTRIBUTE	VALUE	DETAIL	ROLE
Trigger Condition	Close < Low	Close price falls below the Low price for a given ticker.	PREDICATE
Group By	Ticker	Alert evaluates independently per symbol.	PARTITION
Action	Email or Teams	Notification to defined recipient. Body includes ticker, current Close, and event Date.	NOTIFY
~~Behaviour~~	Deduplicate	Alert must not produce duplicate notifications for repeated breaches within the same evaluation window.	SUPPRESS

SECTION 04 · KQL QUERY LIBRARY

Eight queries. Each with a documented business purpose.

The query library is the primary analytical artefact the Trading Domain will consume. Every query must address a defined business requirement, must be validated against live streaming data before registration, and must include a documented business purpose in the MarketDataQueries KQL Queryset.

#	QUERY NAME	BUSINESS REQUIREMENT	TILE
Q1	Latest Close per Ticker	Surface the most recent closing price for each ticker. Current market reference for the trading desk.	T1
Q2	Intra-Day Range per Ticker	Average difference between High and Low per ticker. Proxy for intra-day volatility across the universe.	T2
Q3	Daily Return per Ticker	Average difference between Close and Open per ticker. Measures directional movement and identifies trending instruments.	REF
Q4	Volume Surge Detection	Tickers where maximum volume significantly exceeds average volume. Flags abnormal activity that may signal institutional movement.	T3
Q5	Price Drawdown from Recent High	Percentage decline from each ticker's peak High to latest Close. Surfaces instruments in potential reversal or distress.	REF
Q6	Anomaly Detection	Detect data quality violations: zero or negative prices, negative volume, High below Low. Stops bad data propagating into signals.	DQ
Q7	Pipeline Freshness Check	Time elapsed since most recently ingested tick per symbol. Confirms each monitored ticker is receiving live events.	OPS
Q8	Derived Field Integrity	Validate that derived field calculations produce logically consistent results. Compliance and data integrity check.	DQ

SECTION 05 · DELIVERABLES

Ten artefacts. Each with a defined acceptance condition.

The following deliverables constitute the complete scope. Each has a defined acceptance condition. Partial completion does not satisfy acceptance. All deliverables must be confirmed against the live running platform. Screenshots, exported query results, and audit log records must be retained as evidence.

ID	DELIVERABLE	ACCEPTANCE CONDITION	FORM
D1	Fabric Workspace	VelocityTrade-RTI workspace, F2 capacity or higher, visible in app.fabric.microsoft.com.	PROVISIONED
D2	Eventhouse + KQL DB	VelocityTradeEventhouse with its automatically created KQL Database, both visible with correct naming.	CREATED
D3	StockTicks Schema	All eight fields with correct data types, confirmed in the KQL Database panel prior to Eventstream connection.	SCHEMA
D4	PipelineAuditLog	All four fields. At least one entry logged recording the schema creation event. First audit entry visible via query.	AUDIT INIT
D5	TickIngestionStream	Eventstream with StockMarketSample connected, routing confirmed to StockTicks. Live events visible. Non-zero count.	FLOWING
D6	KQL Query Library	All eight queries built, validated against live data, registered in MarketDataQueries with documented business purpose.	REGISTERED
D7	RTI Dashboard	VelocityTrade Live Market Intelligence with three tiles. Each powered by a live KQL query. Auto-refresh set to 30 seconds.	RENDERING
D8	DrawdownAlert	Activator configured against StockTicks with trigger, grouping, action, message content per Exhibit 8. Activated and tested.	FIRED
D9	Audit Completeness	PipelineAuditLog contains entries for every configuration change across the delivery. No gaps. Every entry has full metadata.	COMPLETE
D10	End-to-End Trace	Documented walkthrough confirming a single tick event can be traced from source through stream, table, query, tile, alert.	TRACED

SECTION 05 · ACCEPTANCE CRITERIA

Seven criteria. All confirmed against the live platform.

Final acceptance requires all criteria to be confirmed simultaneously against the live running platform. Acceptance is binary. Each criterion is either met in full against live data or it is not met. There is no partial credit for components that were working at an earlier point but are not demonstrable at the time of review.

01	PIPELINE CONTINUITY	TickIngestionStream in running state with no error indicators. StockTicks count non-zero and increasing at review.
02	SCHEMA INTEGRITY	StockTicks contains all eight fields per Exhibit 4. PipelineAuditLog contains all four fields per Exhibit 5.
03	QUERY LIBRARY COMPLETENESS	All eight queries present in MarketDataQueries, each with documented business purpose, each returning expected results against live data.
04	DASHBOARD OPERATION	All three tiles rendering live data at review. Auto-refresh active at 30 seconds. Each tile powered by correct query.
05	ALERT VALIDATION	DrawdownAlert active. Threshold breach confirmed. Notification produced and received. Duplicates suppressed within window.
06	AUDIT LOG COMPLETENESS	PipelineAuditLog covers every configuration change. No gaps. Every entry has ChangeTimestamp, Author, ComponentChanged, ChangeDescription.
07	GOVERNANCE COMPLIANCE	Write access verified as restricted to the Eventstream pipeline identity. Failed direct write from an analyst account demonstrable as evidence of access control enforcement.

SECTION 06 · DISCUSSION FRAMEWORK

Eight questions. Each grounded in the pipeline you built.

The following questions test the depth of understanding behind each architectural and operational decision. Answers must be grounded in the specific context of the VelocityTrade pipeline as built, not in generic platform descriptions.

Q1 · ARCHITECTURE OWNERSHIP

The mandate prohibits direct writes to StockTicks. A team member argues writing directly from a notebook during initial setup is faster. What governance, auditability, and operational risks does the direct write approach introduce? How does this boundary connect to MiFID II compliance?

Q2 · SCHEMA DESIGN

StockTicks defines Volume as long rather than integer or real. What is the technical rationale given expected NYSE and NASDAQ volumes? What happens if Volume is defined as integer, and how would the failure manifest in the Volume Surge query results?

Q3 · QUERY ENGINEERING TRADEOFFS

The library derives IntraDayRange and DailyReturn at query time within KQL rather than as stored computed columns. What are the operational tradeoffs between query-time derivation and ingestion-time enrichment? Under what conditions would stored enrichment become preferable?

Q4 · PIPELINE FRESHNESS

Q7 measures elapsed time since most recently ingested event per ticker. At what staleness threshold should this trigger an operational alert? How would you integrate Q7 into the Activator framework to produce an automated freshness breach notification?

Q5 · ANOMALY DETECTION

Q6 flags zero or negative prices, High less than Low, and negative volume. What additional data quality conditions should this query check that are not currently covered? How would even a small number of anomalous records affect the Volume Surge and Price Drawdown query results if left unfiltered?

Q6 · COMPLIANCE RECONSTRUCTION

A regulatory inquiry requires reconstruction of the exact market data state for MSFT at 14:32:17 UTC on a specific automated trade execution date. Using StockTicks and PipelineAuditLog as your evidence base, describe the KQL query logic. Identify any gaps in the current schema that would limit reconstruction precision.

Q7 · SCALABILITY & GOVERNANCE

The trading desk intends to onboard live feeds from Asian equity markets next quarter, sourced from a different provider with a different event schema. What changes to TickIngestionStream, the StockTicks schema, and the PipelineAuditLog governance process would be required to support this expansion without breaking the existing queries or DrawdownAlert?

Q8 · BUSINESS VALUE MEASUREMENT

Now that signal latency has been reduced from up to 45 minutes to sub-second, construct a measurement framework using KQL queries that would allow the quantitative research team to quantify the improvement in signal timing accuracy in the first full quarter of live operation. What baseline data would you need and how would you isolate the latency improvement effect from other sources of return variance?

SECTION 06 · LAB INSTRUCTIONS & RISKS

Phase by phase. Risks named and mitigated.

You are the Lead Data Engineer on the Enterprise Platform Team. All work must be performed within the VelocityTrade-RTI workspace. No external tools, notebooks, or direct database writes are permitted at any stage. Each phase has a defined input, a required operation, and a named output.

PHASE	OPERATION	DETAIL	OUTPUT
WORKSPACE	Provision	VelocityTrade-RTI with F2 capacity or higher. All pipeline components created within this workspace boundary.	WORKSPACE
EVENTHOUSE	Create	VelocityTradeEventhouse. Within its KQL Database, create StockTicks and PipelineAuditLog with the schemas in Exhibits 4 and 5. Both tables before any Eventstream connection.	2 TABLES
EVENTSTREAM	Wire	TickIngestionStream connected to StockMarketSample. Configure routing to StockTicks. Validate non-zero increasing row count.	STREAM
QUERIES	Build	All eight queries in MarketDataQueries with business purpose comment. Each validated against live data before registration.	8 QUERIES
DASHBOARD	Deploy	VelocityTrade Live Market Intelligence with three tiles per Exhibit 7. Auto-refresh at 30 seconds. All tiles rendering live before sign-off.	DASHBOARD
ALERT	Configure	DrawdownAlert per Exhibit 8. Activated and confirmed. Notification received. No duplicates within window.	ALERT
AUDIT	Record	Every configuration change recorded in PipelineAuditLog. Queryable and complete at final review.	LOG
VALIDATION	Trace	Complete end-to-end architecture trace. Every component live and demonstrable. Any gap is an unresolved delivery risk.	PASS

KEY RISKS

Schema mismatch between source and KQL table can land events with corrupted or missing field values without raising an error; define explicit field mappings in the Eventstream destination. KQL query performance against growing event volume requires time-window filters or aggregation queries will degrade. Activator misconfiguration produces either false positives or silence on real breaches; test with live data before sign-off. The Fabric sample source can pause without raising an explicit failure; monitor Q7 actively as the freshness signal.

From batch snapshots to sub-second tick intelligence.

Build the Real-Time Intelligence pipeline that gives VelocityTrade a market data feed it can trade against. Eventstream ingesting StockMarketSample tick by tick. KQL Database with a StockTicks table whose schema is defined before a single event lands. Eight queries registered in a governed library, each validated against live data. A three-tile dashboard refreshing every thirty seconds. An Activator alert that fires when Close falls below Low and suppresses duplicates within the window. One audit log that records every configuration change. One end-to-end trace from sample source to alert.

WHAT YOU WILL DELIVER

Ten artefacts. A provisioned Fabric workspace with F2 capacity. An Eventhouse hosting a KQL Database with StockTicks and PipelineAuditLog. A TickIngestionStream connected to StockMarketSample. An eight-query MarketDataQueries library. A three-tile RTI dashboard with thirty-second refresh. A DrawdownAlert via Activator. A complete PipelineAuditLog. A documented end-to-end trace.

HOW YOU WILL BE EVALUATED

TickIngestionStream is running with non-zero increasing event count. All eight queries return expected results against live data. All three tiles render live with auto-refresh confirmed. DrawdownAlert fires on a real breach and suppresses duplicates. PipelineAuditLog has no gaps. Direct write attempt from an analyst account fails. Eight discussion answers grounded in the live platform, not in documentation.

WHAT THIS PROVES

You can build a real-time ingestion pipeline that is governed from the first tick. You can choose long over integer for a Volume column because you understand NYSE per-tick volumes. You can defend query-time derivation over ingestion-time enrichment as a workload fit decision. You can wire Activator to a freshness check as well as a price-breach check. You can deliver critical trading infrastructure that survives a compliance reconstruction request.

WHAT COMES NEXT

Onboard Asian equity feeds from a different provider with a different schema. Add derived fields as enriched stream attributes for the queries that need them most. Add an Activator alert on Q7 freshness so the trading desk learns of source pauses in seconds. Add row-level security on MarketDataQueries by desk. The pipeline becomes the platform. The strategies finally trade against ground truth.

INSTRUCTOR CONTACT

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